Automatic Category Suggestion for Wikimedia Commons

Hackathon final report | QWERTYBOYS | 11 October 2026 | Code licence: MIT

1. Summary

We built a system that takes a newly uploaded image, plus an optional title, and returns a ranked list of Wikimedia Commons categories for a human to accept or reject. It treats categories as retrievable objects instead of a fixed softmax output. A frozen CLIP encoder (ViT-B/32) feeds four scorers: kNN retrieval over categorised images, text prototypes, a supervised head trained with asymmetric loss, and a title scorer. A logistic-regression fusion and a hierarchy post-process combine them. On 361 held-out images from later uploads, the text-prototype scorer places a correct category first for **81.2%** of images and the fused ranking for **74.5%**; a correct category is in the top five for **95.7%** and **93.8%** of images. One image takes about 92 ms to encode on a laptop CPU and under 1 ms to rank, so no GPU is needed for Commons-scale upload volume. New images and categories are added without retraining.

2. Approach

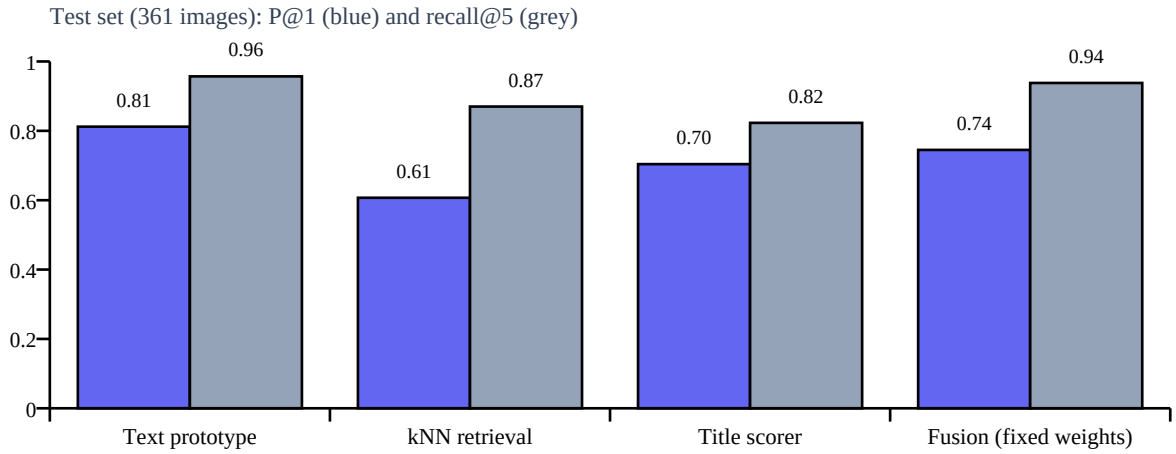
Pipeline: upload, CLIP image embedding, four scorers over candidate categories, calibrated fusion, hierarchy clean-up, then ranked suggestions with the nearest categorised images shown as the explanation. **kNN retrieval** lets the K nearest indexed images vote with softmax-weighted similarity; it needs no retraining and explains itself. **Text prototypes** compare the image with the CLIP text embedding of each category name, which covers categories with few or no examples. **The supervised head** is a linear multi-label layer initialised from the text prototypes and trained with asymmetric loss, which down-weights easy negatives and tolerates missing labels through a probability margin. **The title scorer** compares the file title with category names, which helps for places and events. **Fusion** is a logistic regression over the four scores, fitted on a validation split and applied only to the union of each scorer's top candidates. **Hierarchy:** editors label with the most specific category, so a proposed ancestor is dropped when one of its descendants is also proposed.

3. Data and protocol

Source: files and their categories were collected through the MediaWiki API; all Commons files are freely licensed by policy. Hidden maintenance categories were excluded at query time, and licence, upload-process and author categories were removed with a pattern filter. **Final collection:** 13,584 images and 148 categories, used by the demo application. **Evaluation protocol (applied to the pilot dataset reported below):** the split is by upload time, never random, so the test set holds the newest uploads. The category vocabulary is counted on training images only. Near-duplicates of training images (CLIP cosine similarity above a threshold) are removed from validation and test. The pilot dataset has 35 categories, 1,296 training images in the index and 361 test images. Metrics are per-image precision, recall and F1 at k, plus per-category macro recall split into head, torso and tail categories (thirds by training frequency).

4. Results on the temporal test set (pilot dataset)

Method	P@1	R@3	R@5	F1@3	Macro R@5	Tail R@5
Text prototype (zero-shot)	0.812	0.928	0.957	0.466	0.967	0.958
kNN retrieval	0.607	0.812	0.870	0.408	0.885	0.838
Title text scorer	0.704	0.798	0.823	0.402	0.748	0.848
Fusion (fixed weights, no head)	0.745	0.909	0.938	0.457	0.953	0.922
Fusion + hierarchy	0.745	0.909	0.938	0.457	0.953	0.922



Almost every test image has exactly one in-vocabulary label (P@1 and R@1 are nearly equal), so precision at k above 1 is capped near 1/k (P@5 cannot exceed 0.2). We therefore read P@1 and recall at 3 and 5, which is what a reviewer sees in a suggestion list. **Findings.** (1) With only 35 categories and about 37 training images per category, zero-shot text prototypes are the strongest single scorer, ahead of kNN. (2) The fixed-weight fusion in this run favours kNN, the weakest scorer here, and lands below the text prototype by 6.7 points, more than the sampling margin of about +/-4.5 points on 361 images; the fitted fusion with the trained head is evaluated on degraded images below. (3) The hierarchy step changed no metric, most likely because the vocabulary has almost no categories that are ancestors of one another. (4) Scores are strict: stored Commons labels are incomplete, so a correct suggestion that editors did not tag counts as wrong. No human-reviewed estimate was collected.

Robustness to low image quality

Method	P@1 clean	P@1 degraded	Change	R@5 clean	R@5 degraded	Change
Text prototype	0.812	0.756	-0.056	0.957	0.936	-0.021
kNN retrieval	0.607	0.501	-0.106	0.870	0.796	-0.074
Title scorer (image-independent)	0.704	0.704	0.000	0.823	0.823	0.000

Degraded images were reduced to a quarter of their resolution and re-encoded as low-quality JPEG before embedding, with index and models unchanged. kNN suffers most because its neighbours are found by fine image similarity; text prototypes lose about half as much. For the trained components on degraded images: the supervised head alone reaches P@1 0.698 and R@5 0.918, and the fitted fusion with the head reaches P@1 0.740 and R@5 0.950 (361 images, 35 categories).

5. Efficiency and scalability

Measured on a laptop CPU with no GPU: image-to-embedding latency p50 92 ms and p95 162 ms; ranking (kNN, fusion, hierarchy) p50 0.51 ms and p95 0.64 ms; batched throughput 21.6 images/s, about 19 times the 1.2 images/s needed for 100,000 uploads a day. The exact index of the pilot holds 1,296 vectors in 2.7 MB (512 dimensions, fp32). Scaling arithmetic: 10^8 images at 512 dimensions in fp16 is about 102 GB flat, or about 6.4 GB with 64-byte product quantisation, with HNSW or IVF-PQ lookups in milliseconds. Inference is asynchronous (upload, queue, encode, lookup, store suggestions) and never blocks the upload. No per-category model exists, so adding categories adds no inference cost.

6. Continual learning and human in the loop

Index updates: accepted images are appended to the nearest-neighbour index immediately, with no retraining. **New categories:** a category is usable at once through its text prototype and any example images; the demo lets a reviewer add a missing category, and the head is extended at the next scheduled retrain. **Feedback:** accept and reject events are logged, and rejects are kept as hard negatives, which are more reliable than absent labels. **Monitoring (design):** track acceptance rate and calibration per category family and retrain on drift. Nothing is edited automatically; every suggestion needs human approval.

7. Wikimedia fit, limitations and risks

Fit: open code (MIT), openly released encoder weights, suggestions explained by the nearest categorised images, suggestions limited to content categories, and no identification of individuals. **Limitations:** (1) The evaluation above covers the 35-category pilot; the final 148-category dataset was built afterwards and its evaluation is not included in this report. (2) Both datasets are a tiny fraction of Commons' millions of categories and come from seed topics we chose, so scores are not a forecast for the full site. (3) Strict scoring undercounts correct suggestions. (4) With 361 test images, differences of a few points are within sampling noise. (5) File titles often name the subject; a real upload may have a poor title or none, and the fusion model was trained with titles. (6) Seed categories lean toward well-known subjects, so geographic and cultural coverage was not audited. (7) Events and abstract concepts rely on titles and text prototypes and were not evaluated separately. (8) Everything was measured on one CPU laptop.

8. Where each judging criterion is addressed

Criterion	Evidence
Accuracy on new images	Section 4: temporal split, near-duplicate removal, P@k, R@k, F1, head/torso/tail recall
Category breadth	Text prototypes and title scorer for rare, place and event categories (Sections 2 and 7); not separately measured
Efficiency and scalability	Section 5: latency, throughput, index size, scaling arithmetic
Robustness	Section 4: clean versus degraded test images
Continual learning	Section 6: index append, new categories, feedback with hard negatives
Wikimedia fit and usability	Sections 6 and 7: open licence, explanations, human approval, no person identification